

7824 Zero Road, second floor

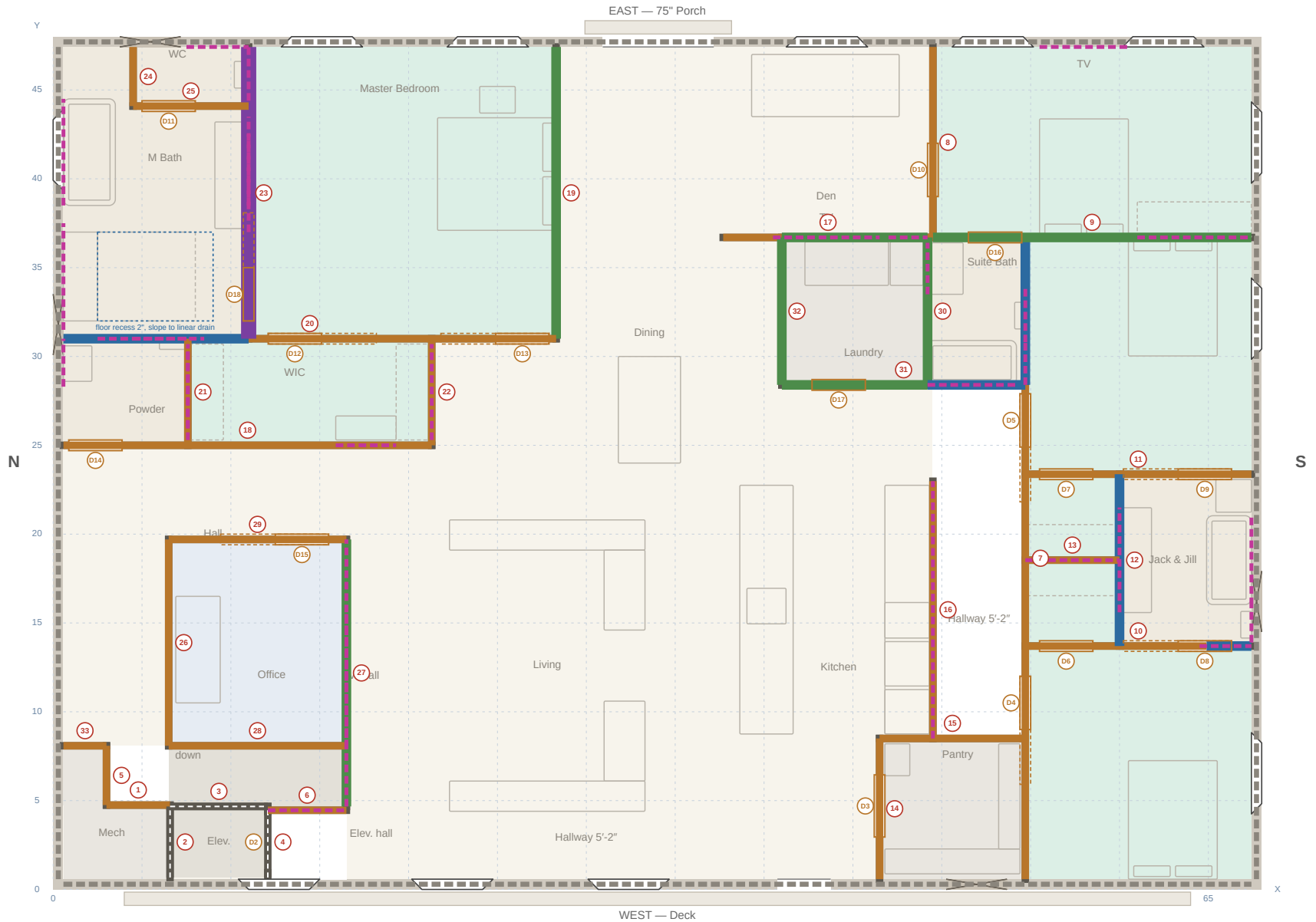
S-1 Framing plan · packet for the framing contractor

Interior partitions on the O3B second floor: wall types, door rough openings, headers, blocking, and the exterior furring as given.

Rev C · 2026-10-01 · Schematic for pricing and coordination, not a stamped construction document. Verify every dimension on site. Coordinates are feet from the north and west brick faces; the plan is drawn with east at the top and north on the left. Wall numbers match the concept drawing. Questions: name the sheet and the tag (for example S-1 W23).

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7824 Zero Road — second floor

Casper, Wyoming · 68' x 48' to brick, faces 7" inside · 9' ceiling · grid 5'
Wall numbers as the floor plan drawing of 2026-09-29

Framing plan

Interior partitions: wall types, door rough openings, headers and blocking

Schematic for pricing and coordination, not a stamped construction document. Verify on site.

S-1

rev C · 2026-10-01

drawn for K. Christianson

Notes

1. All interior walls are non-bearing partitions under a clear-span 9'-0" flat ceiling. Headers: a single flat 2x4 for every opening up to 8' (IRC R602.7.4), cripples at 16" to the double top plate. Pocket doors use the frame maker's header.
2. Ceiling: roof trusses run east-west (along Y) at 24" o.c. Partitions that run east-west sit between trusses and need 2x6 flat blocking at 24" between the bottom chords to catch the top plate; partitions that run north-south cross a truss every 24". Hang every top plate with slip clips (1" gap, no nails through the plate into the chord) to avoid truss uplift cracking.
3. Walls are drawn at 5" (2x4 + 1/2" board each side). A 2x6 wet wall is 6-1/2"; the extra 1-1/2" grows toward the side named in the schedule, so the room on the other side keeps its drawn dimension.
4. Doors: 8'-0" leaves. Swing door RO = leaf + 2" wide × 98-1/2" tall. Pocket door RO = 2 × leaf + 1" wide × 100-1/2" tall; the dashed box shows the pocket side. D3 is a 42" cased opening. D4, D5 (36" pockets) and D8, D9 each have a partition abutting the pocket side of the wall (walls 15, 11 and 12): the abutting wall nails to the pocket frame's reinforced split studs, not through the cavity.
5. Staggered-stud walls (green): 2x6 plates, 2x4 studs alternating faces at 12" o.c. (24" o.c. on each face), R-19 batt woven between, no stud touches both faces. Pipes and the washer box thread between the studs; the dryer gets a recessed dryer box in wall 17. 2x10 wall 23 (purple): 2x10 plates and studs at 16" o.c.; the pocket frame for D18 sits in the bedroom half, the vanity drains and the WC vent in the bath half.
6. Exterior furring (grey dashed): 2x4 @ 24" o.c. in the 7" band, standing 1/2" off the brick as an air gap. Shown in the 3D model; nothing changes there except the blocking listed.
7. Every door has at least 32" clear and 18" latch-side clearance on the pull side per the plan; keep thresholds at 1/2" max (1/4" at the shower, which has none).
8. Curbless shower (X 2'-6" to 8'-0", Y 32' to 37'): recess the floor 2" over the pad and 12" beyond its open side, slope 1/4" per foot to the linear drain on the bench side.
9. Blocking is shown as magenta bands on the face noted. Grab-bar blocking is continuous 2x8 flat with its centre at 34" AFF; vanity blocking 2x10 flat 26"–34" AFF (wall-hung roll-under vanities, 27" clear beneath); TV blocking 3/4" plywood 42"–66" AFF.
10. Sound: R-13 batts in every bedroom and bathroom wall and in walls 7, 9, 17, 19, 20 and 23.
11. Exterior walls: brick shell with the measured 7" furring. New blocking in the furring is shown on the sheet; nothing structural changes there. No water piping in the furring (see P-1).
12. Elevator hoistway walls (2, 3, 4) and the stair opening are as measured; the stair guard and handrails follow IRC R311.7.8 and R312 (34"–38" rail, 36" guard, 4" sphere).
13. Sub-panel P2 sits on the landing face of wall 1 so it stays reachable.

Open items

- Pocket doors D4, D5, D8, D9: each pocket crosses a T-junction (walls 15, 11, 12). Workable with the frame's steel split studs; confirm, or swap D4 and D5 back to swings.
- Door leaf style and hardware: lever handles, 42" leaves for Bedrooms 3 and 4, and whether the elevator-hall side of wall 6 is a full wall or a guard.

- Confirm the wet-wall growth direction on walls 7, 17, 20 and 31 (each steals 1-1/2" from the bath or laundry side).
- 2nd Suite wardrobe: a 2' deep reach-in needs a front wall and doors that are not on the plan; confirm, or keep it as freestanding wardrobes.

Schedules

Wall schedule

Wall	Runs	Length	Framing	2x6 grows	Doors	Notes
W1	X 3'-0"–6'-7" at Y 4'-9"	3'-7"	2x4 partition			Mech room, landing side (X 3' to the hoistway). 3/4" plywood backing 48" × 48" on the landing face for sub-panel P2. The mech room and the vent leg are one L-shaped room with no door.
W2	Y 8"–4'-8" at X 6'-7"	4'-0"	existing hoistway			Hoistway, west.
W3	X 6'-7"–12'-1" at Y 4'-8"	5'-6"	existing hoistway			Hoistway, south, with the 3'-0" car door opening to the elevator hall.
W4	Y 8"–4'-8" at X 12'-1"	4'-0"	existing hoistway		D2 Elevator car door	Hoistway, east.
W5	Y 4'-9"–8'-1" at X 3'-0"	3'-4"	2x4 partition			Mech closet leg (vents).
W6	X 12'-1"–16'-6" at Y 4'-6"	4'-5"	2x4 partition			Elevator hall to stair. Full height; stair side gets the handrail blocking.
W7	Y 7"–36'-8" at X 54'-8"	36'-1"	2x4 partition (7"–28'-5") + 2x6 wet wall (28'-5"–36'-8")	west	D4 Bedroom 3, D5 Bedroom 4	Wing wall, 36'-1". 2x6 from Y 28'-5" to 36'-8" for the suite-bath toilet vent; grows into the bath.
W8	Y 36'-8"–47'-5" at X 49'-6"	10'-9"	2x4 partition		D10 2nd Suite	2nd Suite west wall.
W9	X 49'-6"–67'-5" at Y 36'-8"	17'-11"	2x6 plate, staggered 2x4 studs	north	D16 Suite bath	2nd Suite / Bedroom 4. Staggered studs for sound; headboards both sides. Grows into Bedroom 4.
W10	X 54'-8"–67'-5" at Y 13'-8"	12'-9"	2x4 partition (54'-8"–64'-11") + 2x6 wet wall (64'-11"–67'-5")	south	D6 Closet 3, D8 Jack & Jill	Bedroom 3 / closet 3 / Jack & Jill. 2x6 at the tub end (X 64'-11" to the brick) for the tub valve; grows into the bath.
W11	X 54'-8"–67'-5" at Y 23'-4"	12'-9"	2x4 partition		D7 Closet 4, D9 Jack & Jill	Bedroom 4 / closet 4 / Jack & Jill.
W12	Y 13'-8"–23'-4" at X 60'-0"	9'-8"	2x6 wet wall	west		Jack & Jill vanity wall: two lav drains and stack S4. Grows into closets 3 and 4.

Wall	Runs	Length	Framing	2x6 grows	Doors	Notes
W13	X 54'-8"– 60'-0" at Y 18'-6"	5'-4"	2x4 partition			Between closets 3 and 4.
W14	Y 7"–8'-6" at X 46'-6"	7'-11"	2x4 partition		D3 Pantry	Pantry west wall, hallway side.
W15	X 46'-6"– 54'-8" at Y 8'-6"	8'-2"	2x4 partition			Pantry / kitchen.
W16	Y 8'-6"–23'-0" at X 49'-6"	14'-6"	2x4 partition			Kitchen wall run. Continuous 3/4" plywood backing 0–96" AFF on the kitchen face for tall units, oven tower and hood; stack S2 (2") inside.
W17	X 37'-8"– 49'-6" at Y 36'-8"	11'-10"	2x4 partition (37'-8"–41'-0") + 2x6 plate, staggered 2x4 studs (41'-0"–49'-2") + 2x4 partition (49'-2"–49'-6")	north		Den / laundry. Staggered-stud sound wall along the laundry (washer box and laundry sink thread between the studs; recessed dryer box); grows into the laundry. TV backing on the den face X 40'-6" to 46'-6".
W18	X 7"–21'-4" at Y 25'-0"	20'-9"	2x4 partition		D14 Powder	Hall / powder and walk-in.
W19	Y 31'-0"– 47'-5" at X 28'-4"	16'-5"	2x6 plate, staggered 2x4 studs	west		Master bedroom east wall; headboard wall. Staggered studs for sound; grows into the bedroom.
W20	X 7"–28'-4" at Y 31'-0"	27'-9"	2x6 wet wall (7"–11'-0") + 2x4 partition (11'-0"–28'-4")	south	D12 WIC, D13 Master bedroom	Master bath and bedroom south wall. 2x6 from the north face to X 11'-0" for the powder toilet, shower valve and stack S1; grows into the powder room and walk-in.
W21	Y 25'-0"– 31'-0" at X 7'-7"	6'-0"	2x4 partition			Powder / walk-in.
W22	Y 25'-0"– 31'-0" at X 21'-4"	6'-0"	2x4 partition			Walk-in east wall. Rod and shelf backing.
W23	Y 31'-0"– 47'-5" at X 11'-0"	16'-5"	2x10 wall	east	D18 Master bath	Bath / bedroom wall, 2x10: pocket door D18 at the south end, the double vanity drains, the WC toilet vent and the TV wiring share it. Grows into the bedroom. TV backing on the bedroom face.
W24	Y 44'-1"– 47'-5" at X 4'-6"	3'-4"	2x4 partition			WC west wall. The tub beyond is freestanding; no valve in this wall.
W25	X 4'-6"–11'-0" at Y 44'-1"	6'-6"	2x4 partition		D11 WC	WC door wall.
W26	Y 8'-1"–19'-8" at X 6'-6"	11'-7"	2x4 partition			Office west wall.

Wall	Runs	Length	Framing	2x6 grows	Doors	Notes
W27	Y 4'-8"—19'-8" at X 16'-6"	15'-0"	2x6 plate, staggered 2x4 studs	west		Office and stair east wall. Staggered studs for sound; grows into the office and stair. TV backing on the living face Y 8' to 16'; handrail blocking on the stair face.
W28	X 6'-6"—16'-6" at Y 8'-1"	10'-0"	2x4 partition			Stair / office.
W29	X 6'-6"—16'-6" at Y 19'-8"	10'-0"	2x4 partition		D15 Office	Office north wall with the office door.
W30	Y 28'-5"— 36'-8" at X 49'-2"	8'-4"	2x6 plate, staggered 2x4 studs	west		Laundry / suite bath: staggered-stud sound wall; the suite vanity drain and vent stack S3 thread between the studs. Grows into the laundry.
W31	X 41'-0"— 54'-8" at Y 28'-5"	13'-8"	2x6 plate, staggered 2x4 studs (41'-0"— 49'-2") + 2x6 wet wall (49'-2"—54'-8")	north	D17 Laundry	Hall / laundry and suite bath. Staggered studs along the laundry, 2x6 behind the suite tub for its valve; grows into the laundry and bath.
W32	Y 28'-5"— 36'-8" at X 41'-0"	8'-4"	2x6 plate, staggered 2x4 studs	east		Laundry west wall, dining side, moved north 1' to X 41'. Staggered studs for sound; grows into the laundry.
W33	X 7"—3'-0" at Y 8'-1"	2'-5"	2x4 partition			Vent leg, passage side: a wall where door D1 was.

Door schedule

Door	Room	From	Wall	Leaf	Type	Rough opening	Header
D2	Elevator car door	elevator hall	W4	3'-0" car door	hoistway	by elevator contractor	—
D3	Pantry	hallway	W14	3'-6" cased opening, 8'-0"	opening	44" × 98.5"	flat 2x4
D4	Bedroom 3	wing hallway	W7	3'-0" × 8'-0"	pocket toward lower Y	73" × 100.5"	frame kit
D5	Bedroom 4	wing hallway	W7	3'-0" × 8'-0"	pocket toward lower Y	73" × 100.5"	frame kit
D6	Closet 3	Bedroom 3	W10	3'-0" × 8'-0"	swing	38" × 98.5"	flat 2x4
D7	Closet 4	Bedroom 4	W11	3'-0" × 8'-0"	swing	38" × 98.5"	flat 2x4
D8	Jack & Jill	Bedroom 3	W10	3'-0" × 8'-0"	pocket toward lower X	73" × 100.5"	frame kit
D9	Jack & Jill	Bedroom 4	W11	3'-0" × 8'-0"	pocket toward lower X	73" × 100.5"	frame kit
D10	2nd Suite	den	W8	3'-0" × 8'-0"	swing	38" × 98.5"	flat 2x4
D11	WC	master bath	W25	3'-0" × 8'-0"	swing	38" × 98.5"	flat 2x4
D12	WIC	master bedroom	W20	3'-0" × 8'-0"	pocket toward higher X	73" × 100.5"	frame kit
D13	Master bedroom	open corner	W20	3'-0" × 8'-0"	pocket toward lower X	73" × 100.5"	frame kit
D14	Powder	hall	W18	3'-0" × 8'-0"	swing	38" × 98.5"	flat 2x4
D15	Office	hall	W29	3'-0" × 8'-0"	pocket toward lower X	73" × 100.5"	frame kit

Door	Room	From	Wall	Leaf	Type	Rough opening	Header
D16	Suite bath	2nd Suite	W9	3'-0" × 8'-0"	swing	38" × 98.5"	flat 2x4
D17	Laundry	dining	W31	3'-0" × 8'-0"	swing	38" × 98.5"	flat 2x4
D18	Master bath	master bedroom	W23	3'-0" × 8'-0"	pocket toward higher Y	73" × 100.5"	frame kit

Blocking schedule

Tag	Wall	Span	Blocking	For
B1	W20	2'-6"—8'-6"	grab bar: 2x8 flat, centre 34" AFF	master shower, back wall (bath face)
B2	W20	4'-6"—8'-0"	grab bar: 2x8 flat, centre 34" AFF	powder toilet, rear (powder face)
B3	W21	26'-0"—30'-6"	grab bar: 2x8 flat, centre 34" AFF	powder toilet, side (powder face)
B4	W23	44'-5"—47'-5"	grab bar: 2x8 flat, centre 34" AFF	WC toilet, rear (WC face)
B5	W23	37'-0"—43'-6"	wall-hung vanity: 2x10 flat, 26"—34" AFF	master double vanity, roll-under (bath face)
B6	W23	38'-6"—42'-6"	TV: 3/4" ply 24" tall, 42"—66" AFF	master bedroom TV (bedroom face)
B7	W7	30'-10"—33'-10"	grab bar: 2x8 flat, centre 34" AFF	suite toilet, rear (bath face)
B8	W31	49'-2"—54'-2"	grab bar: 2x8 flat, centre 34" AFF	suite tub, long side (bath face)
B9	W7	28'-5"—30'-11"	grab bar: 2x8 flat, centre 34" AFF	suite tub, end (bath face)
B10	W30	33'-6"—36'-5"	wall-hung vanity: 2x10 flat, 26"—34" AFF	suite vanity, roll-under (bath face)
B11	W17	47'-0"—49'-2"	wall-hung vanity: 2x10 flat, 26"—34" AFF	laundry sink (laundry face)
B12	W17	40'-6"—46'-6"	TV: 3/4" ply 24" tall, 42"—66" AFF	den TV (den face)
B13	W27	8'-0"—16'-0"	TV: 3/4" ply 24" tall, 42"—66" AFF	living TV (living face)
B14	W12	15'-7"—21'-6"	wall-hung vanity: 2x10 flat, 26"—34" AFF	Jack & Jill double vanity, roll-under (bath face)
B15	W10	64'-6"—67'-5"	grab bar: 2x8 flat, centre 34" AFF	Jack & Jill toilet, side (bath face)
B16	W16	8'-6"—23'-0"	3/4" ply backing 0"—96" AFF	kitchen tall units, oven, hood (kitchen face)
B17	W18	15'-11"—19'-4"	rod and shelf: 2x6 flat at 66" and 80" AFF	linen shelves (walk-in face)
B18	W22	25'-4"—30'-8"	rod and shelf: 2x6 flat at 66" and 80" AFF	walk-in rods
B19	W21	25'-4"—30'-8"	rod and shelf: 2x6 flat at 66" and 80" AFF	walk-in rods (walk-in face)
B20	W13	54'-8"—60'-0"	rod and shelf: 2x6 flat at 66" and 80" AFF	closets 3 and 4, both faces
B21	W9	61'-0"—67'-5"	rod and shelf: 2x6 flat at 66" and 80" AFF	2nd Suite wardrobe (suite face)
B22	W27	4'-8"—19'-8"	handrail: 2x6 flat at 34" AFF	stair, open-side handrail
B23	W6	12'-1"—16'-6"	handrail: 2x6 flat at 34" AFF	stair, elevator-hall side
B24	furring	Y 31'-6"—37'-6"	grab bar: 2x8 flat, centre 34" AFF	master shower bench wall (north)
B25	furring	X 7'-6"—11'-0"	grab bar: 2x8 flat, centre 34" AFF	WC toilet, side (east)

Tag	Wall	Span	Blocking	For
B26	furring	Y 28'-4"–31'-0"	wall-hung vanity: 2x10 flat, 26"–34" AFF	powder lav, roll-under (north)
B27	furring	Y 13'-11"–16'-0"	grab bar: 2x8 flat, centre 34" AFF	Jack & Jill toilet, rear (south)
B28	furring	Y 16'-0"–21'-0"	grab bar: 2x8 flat, centre 34" AFF	Jack & Jill tub, long side (south)
B29	furring	X 55'-6"–60'-6"	TV: 3/4" ply 24" tall, 42"–66" AFF	2nd Suite TV (east)
B30	furring	Y 38'-6"–44'-6"	tub filler: floor-mounted, supply from below	master tub (north)

Bill of materials

Bill of materials

Rough-order quantities from the drawing geometry, for pricing. The contractor takes off the final list.

Item	Qty	Notes
Studs 2x4 × 9'	205	16" o.c. plus ends, kings and jacks, +10%
Studs 2x6 × 9'	38	16" o.c. plus ends, kings and jacks, +10%
Studs 2x4 × 9' (staggered)	103	16" o.c. plus ends, kings and jacks, +10%
Studs 2x10 × 9'	15	16" o.c. plus ends, kings and jacks, +10%
Plates 2x4, bottom + double top	619 lf	pressure-treated bottom plate not needed on a wood floor
Plates 2x6, bottom + double top	356 lf	pressure-treated bottom plate not needed on a wood floor
Plates 2x10, bottom + double top	49 lf	pressure-treated bottom plate not needed on a wood floor
Exterior furring 2x4 × 9' @ 24"	136	plus 2x4 plates 464 lf
Flat 2x4 headers with cripples	8	doors D3, D6, D7, D10, D11, D14, D16, D17
Pocket door frame kits, 36" × 96"	8	doors D4, D5, D8, D9, D12, D13, D15, D18; D4, D5 36" (2x4 wall), D18 for a 2x10 wall
Door leaves 8'-0", solid core, lever hardware	15	D4, D5 36"; D3 is a cased opening; D2 by the elevator contractor
2x8 grab-bar blocking	47 lf	centre 34" AFF
2x10 vanity blocking	20 lf	26"–34" AFF
3/4" plywood backing (TV, kitchen wall, panel)	6 sheets	4 × 8 sheets
1/2" gypsum board, interior partitions	211 sheets	4 × 8, both faces, +10%; exterior furring board extra: 72 sheets
R-13 / R-19 sound batts	2170 sq ft	bedroom and bath walls, staggered walls R-19
Slip clips for the top plates (if trusses above)	177	one per 2'
Shower floor recess framing	1	2" drop over 6'-6" × 5' plus 12" apron

Quantities are rough-order take-offs from the drawing geometry, for pricing. The contractor takes off the final list.